Ozge Shera Kuddar, Ph.D.

Bioinformatics Scientist

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Summary

Specialized in probabilistic modeling, prediction with Bayesian inference, bioinformatics pipeline development for the analysis of microbial and environmental sequencing data.

- Led a ~$500K funded research project to develop innovative, application-oriented statistical and probabilistic models resulting in three publications.
  - Related skills: Experimental design, measurement bias detection and mitigation models, forecasting/ prediction models, softmax regression to evaluate classifier performance, and linear / nonlinear regression models.
- Demonstrated experience building **ML project**, including data preprocessing, model development, cross validation and performance assessment, with code available on GitHub.
- **Builder of AI-powered tools for scientific and personal productivity:** Created Open-access tools using privacy-first GenAI - Ollama with open-weight models, including Llama and Qwen. Learning new skills in agentic tools such as LangGraph, LangChain to automate and accelerate the research.
  - Related tools: VS Code Copilot-like ghost-text assistant, AI-powered coding sidebar, LinkedIn job-search assistant, GitHub traffic tracker, and Pomodoro productivity app.
- Proficient in Python, R, and Bash on HPC and AWS; experienced with Git and GitHub version control and familiar with Docker and Nextflow.

Skills

Computing	Python, R, Julia, Bash, Git, Slurm, SAS, SRA, scikit-learn, Biopython, DADA2, QIIME2, NumPy, pandas, Matplotlib, Seaborn, bioinformatics pipeline development, high-performance computing (HPC), cloud computing (AWS), Ollama, Tampermonkey, VSCode.
Statistical	Probabilistic modeling, prediction with Bayesian inference, machine learning, curve fitting (exponential distance-decay modeling), hypothesis testing, analysis of variance, descriptive statistics, correlation and regression, multivariate analysis, parametric and nonparametric tests.
Bioinformatics	DNA metabarcoding, amplicon sequencing, metagenomics, short read and long read sequencing, compositional data analysis, differential abundance analysis, measurement bias, experimental design, taxonomic classification, reference database generation and curation, microbiome and ecological community analysis, diversity and divergence metrics, large-scale data integration and visualization.

Education

PhD, Bioinformatics, NC State University, USA

MBA, Technology, Entrepreneurship & Commercialization, NC State University, USA

MSc, Biotechnology, Technische Hochschule Mannheim, Germany

BEng, Bioengineering, Ege University, Türkiye

Recent Work Experience

2026–Present **Research Scholar, NC State University, USA**

2021–2026 **Graduate Research Assistant, NC State University, USA**

- Expertise: Forecasting and Probabilistic Modeling, Bayesian inference, hypothesis testing, database creating and benchmarking, bioinformatics for microbiome and environmental sequencing data and measurement bias.

Fall 2025 **Teaching Assistant, Microbiome Analysis, NC State University, USA**

- Recitation Lectures: Large Language Models (LLM) and coding; Measurement bias Assisted graduate students without analytical backgrounds in correctly applying statistical methods and bioinformatics pipelines for microbiome data analysis.

Spring 2024 **Internship, Raleigh Biosciences, USA**

- Designed go-to-market and monetization strategies for a single-cell & spatial transcriptomics venture and prepared strategic reports assessing angel investors and venture capital funding options.

2018–2021 **Laboratory Research Analyst I, Duke University School of Medicine, USA**

- Microbial genome engineering (CRISPR), microbiome analysis, host–microbiome interactions, research on *Akkermansia muciniphila* and *Chlamydia trachomatis*.
- Computational analysis of phage sequence databases to identify CRISPR-PAM motifs in *Akkermansia* species.

Selected Publications

In progress

- **Kuddar, O. S.,** Callahan, B. J., Johnson J.W., Bernhardt, C., & Meiklejohn, K. A. Evaluation of DNA Metabarcoding for Characterizing Pollen in Surface Soil Samples with Potential Forensic Applications

2026

- **Kuddar, O. S.,** Meiklejohn, K. A., & Callahan, B. J. (2026). Generating, curating, and evaluating *trnL* reference sequence databases: Benchmarking OBITools3/ecoPCR, RESCRIPt, and MetaCurator. DOI: 10.64898/2026.04.07.717010

2026

- Spencer, P. N., Brown, M. E., Smith, E. P., ... **Kuddar, O. S.,** ... Lau, K. S. (2026). Pathobiont-triggered induction of goblet cell response drives regional susceptibility to Inflammatory Bowel Disease. Journal of Clinical Investigation. DOI: 10.1172/jci201729

2025

- **Kuddar, O. S.,** Callahan, B. J., Bernhardt, C., & Meiklejohn, K. A. (2025). Practical and cost-effective method for the isolation of pollen grains from various sources. Acta Palaeobotanica, 122–129. DOI: 10.35535/acpa-2025-0004

2022

- Dolat, L., Carpenter, V. K., Chen, Y.-S., Suzuki, M., Smith, E. P., **Kuddar, O.,**& Valdivia, R. H. (2022). Chlamydia repurposes the actin-binding protein EPS8 to disassemble epithelial tight junctions and promote infection. Cell Host & Microbe, 30(12), 1685-1700.e10. DOI: 10.1016/j.chom.2022.10.013